



Department of CSE, PES University

UE24CS341A: SOFTWARE ENGINEERING

5TH SEMESTER, ACADEMIC YEAR 2026-27

LAB 1: REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

DATE: 19-08-2026

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ASSIGNMENT WORK

Objective: From a short, instructor-provided scenario, elicit key functions and constraints; write clear, verifiable functional (FR) and nonfunctional (NFR) requirements; then translate them into a UML use-case diagram and one use-case flow.

Final Deliverables

1. Requirements Table (Word/Excel): o Exactly five FRs, out of which two are given (FR-001) and two NFRs out of which one is already given. (NFR-001), each with:

1. Req ID
2. Type (Functional/Nonfunctional)
3. Description (“The system shall...”)
4. Priority (High/Medium/Low)
5. Acceptance Criteria (measurable pass/fail)
6. Rationale (short justification)

2. UML Use-Case Diagram (PDF), showing all actors, use cases, and at least one «include» or «extend» relation.

3. Use-Case Flow Document (Word, one page):

- o For one key use case (e.g., “Place Order”), provide:
 - Main Success Scenario (step-by-step).
 - Alternate Flow, at least one (if payment is declined, for example).

SOLUTION: -

Problem Understanding: -

The **Podcast Guest Scheduling & Outline Builder** is a media production system designed to simplify the process of organizing podcast interviews. It allows podcast hosts to share their available time slots, enables guests to book suitable interview slots and submit their discussion topics and biography links, and helps generate structured production notes for the episode.

Identified Actors: -

The primary actors of the system are the **Podcast Guest** and the **Show Host**. The Podcast Guest books an available interview slot and provides the required discussion details, while the Show Host manages interview availability and uses the collected information for planning the podcast episode.

System Functionality: -

The system provides functionalities such as **managing host availability, booking interview slots, collecting guest talking points and biography links, sending booking/calendar information, and generating a structured episode outline or production sheet**. These functions help organize the complete podcast interview preparation process in a simple and systematic manner.



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Task 1: Requirements Table: -

Req. ID	Type	Description	Priority	Acceptance Criteria	Rationale (Short)	Comments
FR-001	Functional	The system shall allow podcast guests to select and book an available interview time slot.	High	The booking succeeds only when the selected slot is available; blocked or already-booked slots cannot be booked.	Prevents scheduling conflicts and enables guest booking.	—
FR-002	Functional	The system shall allow the Show Host to create, update, and block available interview time slots.	High	Any slot created, updated, or blocked by the host is correctly reflected in the guest's available-slot list.	Allows the host to control interview availability.	—
FR-003	Functional	The system shall allow guests to submit structured discussion points and biography links for the podcast episode.	High	Submitted talking points and biography links are saved and displayed with the correct guest booking.	Collects required information for episode preparation.	—
FR-004	Functional	The system shall send a calendar invitation after a guest successfully books an interview slot.	Medium	A calendar invitation containing the confirmed date and time is generated after successful booking.	Keeps the host and guest informed about the scheduled interview.	—
FR-005	Functional	The system shall generate a structured run-of-show episode outline using the submitted guest information and discussion topics.	High	The generated episode outline contains the guest details and submitted discussion topics in a structured format.	Helps the host prepare and organize the podcast episode.	—
NFR-001	Nonfunctional – Performance	The system shall generate the formatted PDF production sheet in under 1 second .	High	Performance testing confirms that the PDF production sheet is generated within 1 second under the defined test load.	Ensures fast generation of production notes.	—
NFR-002	Nonfunctional – Security	The system shall restrict host-specific scheduling and episode-management functions to authorized users.	High	Unauthorized users attempting to access host-management functions are denied access in all test cases.	Protects scheduling and episode information from unauthorized changes.	—



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Requirements Explanation

The requirements of the **Podcast Guest Scheduling & Outline Builder** are classified into **Functional Requirements (FRs)** and **Non-Functional Requirements (NFRs)**. Functional requirements define **what the system should do**, such as scheduling interviews, collecting guest information, and generating episode outlines. Non-functional requirements define **how effectively the system should operate**, focusing on qualities such as performance and security. Each requirement also contains a priority, measurable acceptance criteria, and rationale so that it can be properly tested and verified.

FR-001 – Interview Slot Booking: This requirement allows the Podcast Guest to view and book an available interview time slot. The system ensures that already booked or blocked slots cannot be selected, thereby avoiding scheduling conflicts.

FR-002 – Host Availability Management: This requirement enables the Show Host to create, modify, or block interview time slots. Any changes made by the host should be immediately reflected in the list of slots available to guests.

FR-003 – Guest Information Submission: This requirement allows the Podcast Guest to submit structured discussion points and biography links after scheduling the interview. The submitted information is associated with the respective guest booking and can be used for episode preparation.

FR-004 – Calendar Invitation: This requirement ensures that a calendar invitation containing the confirmed interview date and time is generated after successful booking. It helps maintain clear scheduling information for the podcast interview.

FR-005 – Run-of-Show Outline Generation: This requirement enables the system to generate a structured run-of-show episode outline using the guest details and submitted discussion topics. It helps the Show Host organize and prepare the podcast episode efficiently.

NFR-001 – Performance: This requirement focuses on the performance of the system. The formatted PDF production sheet should be generated in **less than one second**, ensuring quick access to the required production information.

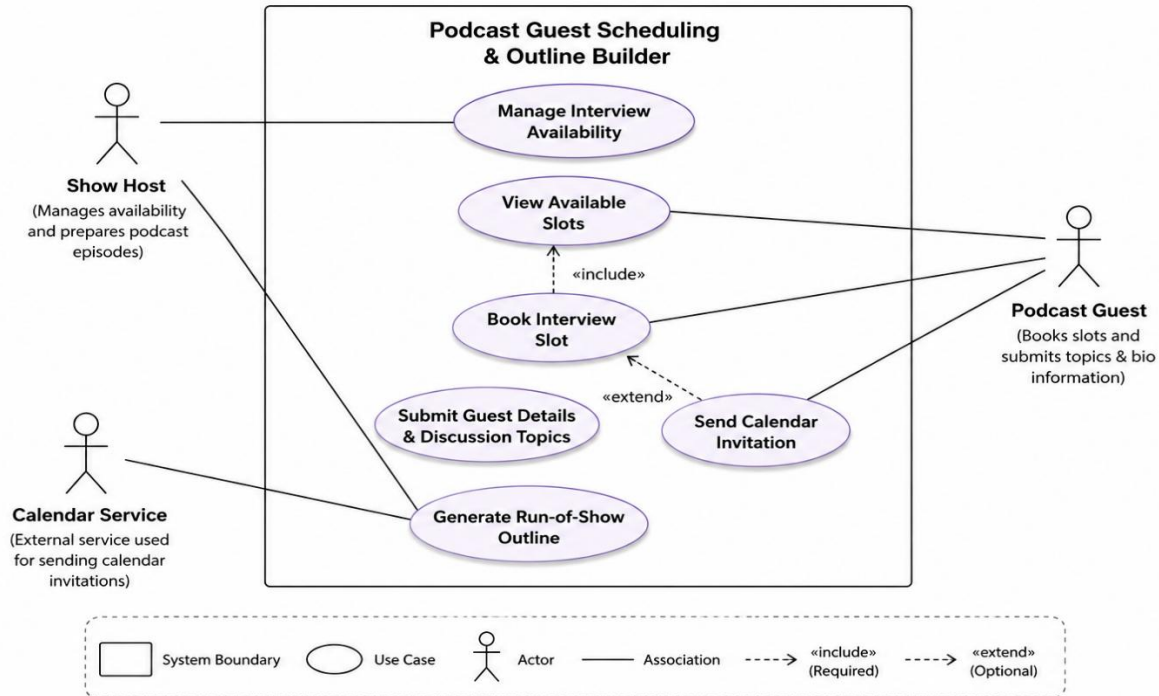
NFR-002 – Security: This requirement focuses on protecting the system from unauthorized access. Host-specific scheduling and episode-management functions should be accessible only to authorized users, preventing unauthorized modification of podcast information.

Identified Actors and Use Cases

Based on the requirements, the main actors identified are the **Podcast Guest, Show Host, and Calendar Service**. The Podcast Guest interacts with the system to view and book available slots and submit discussion details. The Show Host manages interview availability and uses the collected information for episode preparation, while the Calendar Service supports the generation of interview calendar invitations. The major use cases identified are **Manage Availability, View Available Slots, Book Interview Slot, Submit Guest Outline, Send Calendar Invitation, and Generate Run-of-Show Outline**. These actors and use cases form the basis for developing the UML Use-Case Diagram.

Task 2: UML Use-Case Diagram

UML USE CASE DIAGRAM Podcast Guest Scheduling & Outline Builder



UML Use-Case Diagram Explanation

The UML Use-Case Diagram represents the interactions between the **Podcast Guest Scheduling & Outline Builder** and its three main actors: **Show Host**, **Podcast Guest**, and **Calendar Service**. The system boundary contains the major functionalities provided by the application and shows how each actor interacts with them.

Show Host: The Show Host manages the interview availability and prepares the podcast episode. The host can manage available interview slots and generate the final run-of-show outline using the information collected from the guest.

Podcast Guest: The Podcast Guest interacts with the system to view available interview slots, book a suitable slot, and submit guest details and discussion topics required for the podcast episode.

Calendar Service: The Calendar Service represents an external service used for handling calendar-related operations. It supports the system in sending the calendar invitation associated with the scheduled podcast interview.

Manage Interview Availability: This use case allows the Show Host to manage the available dates and time slots for podcast interviews so that guests can select from valid scheduling options.

View Available Slots: This use case allows the Podcast Guest to view the interview slots currently available for booking.

Book Interview Slot: This use case enables the Podcast Guest to select and confirm an available interview slot. It has an «include» relationship with **View Available Slots**, since checking available slots is required as part of the booking process.

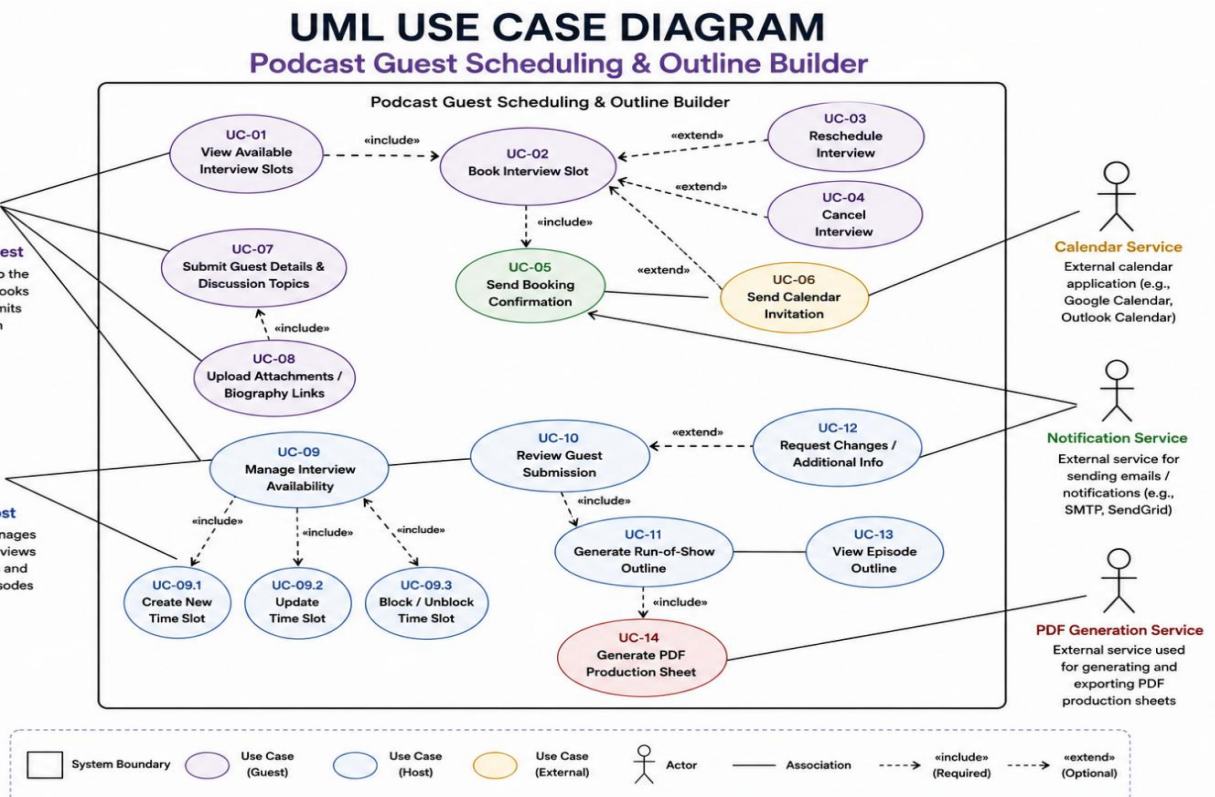
Submit Guest Details & Discussion Topics: This use case allows the Podcast Guest to provide the required guest information, biography details, and discussion topics that will be used for planning the podcast episode.

Send Calendar Invitation: This use case handles the generation and sending of the interview calendar invitation after a successful booking. It is represented using an «extend» relationship with **Book Interview Slot**, as this action occurs after the booking condition has been successfully satisfied.

Generate Run-of-Show Outline: This use case allows the Show Host to generate a structured episode outline using the collected guest details and discussion topics. The generated outline assists the host in organizing and preparing the podcast episode.

Overall, the diagram provides a clear representation of the system boundary, actors, use cases, associations, and the required «include» and «extend» relationships, showing how the major functions of the podcast scheduling and episode preparation system interact.

THE COMPLEX UML Use-Case Diagram (EXTRA)





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TASK 3: Use-Case Flow Specification

Use Case ID: UC-02

Use Case Name: Book Interview Slot

Primary Actor: Podcast Guest

Supporting Actor: Calendar Service

Purpose: To allow a Podcast Guest to select and successfully book an available interview time slot with the Show Host.

Preconditions

1. The Show Host has created and published available interview time slots.
2. The Podcast Guest has accessed the podcast scheduling system.
3. At least one valid interview slot is available for booking.

Postconditions

1. The selected interview slot is successfully reserved for the Podcast Guest.
2. The booked slot is marked as unavailable for other guests.
3. The booking information is stored by the system.
4. A booking confirmation and calendar invitation are generated for the scheduled interview.

Main Success Scenario

1. The Podcast Guest opens the interview scheduling page.
2. The system displays all currently available interview time slots.
3. The Podcast Guest selects a suitable date and time.
4. The system verifies that the selected slot is still available and not blocked by the Show Host.
5. The system displays the selected interview details for confirmation.
6. The Podcast Guest confirms the selected interview slot.
7. The system reserves the selected slot and records the booking details.
8. The booked slot is removed from the list of available slots to prevent duplicate booking.
9. The system generates a booking confirmation.
10. The Calendar Service generates the calendar invitation containing the confirmed interview date and time.
11. The system displays a successful booking message to the Podcast Guest.
12. The use case ends successfully.

Alternate Flow – Selected Slot Unavailable

4a. Selected Interview Slot is No Longer Available

1. During availability verification, the system detects that the selected slot has already been booked or blocked by the Show Host.
2. The system does not create the booking.
3. The Podcast Guest is informed that the selected interview slot is unavailable.
4. The system refreshes and displays the currently available interview slots.
5. The Podcast Guest selects another available slot and the process continues from **Step 3** of the Main Success Scenario.



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SECTION: F

THANK YOU